INCH-POUND
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MILITARY SPECIFICATION

CORD, ELASTIC, NYLON OR POLYESTER

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for elastic cord encased within a braided nylon or polyester cover.

1.2 Classification. The cord will be available in the following types, as specified.

1.2.1 Types.

Type I	General purpose cord
Type II	Lacing cord
Type III	Nylon elastic cord

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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AMSC N/A FSC 8305

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9 Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 192 - Weather Resistance of Textiles: Sunshine Arc-Lamp Exposure with and without Wetting

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org)

ASTM INTERNATIONAL

ASTM D1423 - Standard Test Method for Twist in Yarns by Direct Counting
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric

(Copies of this document are available online at https://www.astm.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cord shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The finished cord shall consist of a core (see 3.7.1) encased by a braided cover (see 3.7.2).

3.5 Materials. Cords shall be constructed from the materials specified for the type (see 1.2) being produced.

3.5.1 Yarn. The yarn used in the production of the cover of Type I and Type II cord and the overwrap of the elastic strands in Type II cords shall be either nylon or textured polyester. The yarn used in the production of the cover of Type III cord shall be 840 denier nylon.

3.5.2 Elastic strands. The elastic strands shall be made of heat-resistant, compounded natural rubber, synthetic rubber, or a mixture thereof. The strands shall be of the gauge specified in Table I.

3.6 Color. When dyed cord is specified, the yarns shall be dyed before braiding.

3.6.1 Visual shade matching. Unless otherwise specified herein or in the contract or purchase order, the color of the finished cord shall match the standard sample (see 6.4) when tested as specified in 4.7.

3.6.2 Colorfastness. When tested as specified in Table V, the finished cord shall show colorfastness to weathering and laundering with a minimum rating of “3-4” and colorfastness to crocking (wet and dry) with a minimum rating of “3.5”.

3.6.7 Construction. The nylon or polyester yarns shall be tightly braided around multiple elastic strands to form a cord conforming to the physical and dimensional requirements specified in Table I, when tested as specified in 4.6 and Table V.

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TABLE I. Physical properties and construction.

Type	Diameter (inches, ± 1/32)	Weight (ounces per linear yard) (min)	Elastic Strands (min)	Number of Carriers	Picks per inch (min)	Rubber gauge	Elongation (percent) 3/
I	7/32	0.70	35	16 (1 end per carrier)	44	24 – 38 1/	90 - 110
II	3/16	0.20	7	16 (1 end per carrier) or 8 (2 ends per carrier)	24	36 2/ (maximum fineness)	115 – 140
III	1/8	0.29	24	16 (1 end per carrier)	48	39 max	65 - 75

1/ Individual strand width between 0.0263 (38 gauge) and 0.0417 (24 gauge) inches.
2/ Individual strand width of 0.0278 (±0.0001) inches. Strands measured without wrapping.
3/ Tested before and after exposure to heat and pressure (see Table V).

3.7.1 Core.

3.7.1.1 Type I. The core of the Type I cord shall consist of a minimum of 35 unwrapped elastic strands (see 3.5.2) grouped together and encased in the outer cover (see 3.7.2)

3.7.1.2 Type II. The core of the Type II cord shall consist of seven (7) elastic strands. Each strand shall be wrapped individually with yarn (see 3.5.1); first with four ends in an S-twist and overwrapped with one end in a Z-twist.

3.7.1.3 Type III. The core of the Type III cord shall consist of a minimum of 24 elastic strands (see 3.5.2) grouped together and encased in the outer cover (see 3.7.2)

3.7.2 Braided cover.

3.7.2.1 Type I. The outer braid of the Type I cord shall be braided over the core (see 3.7.1.1) so as to give a “two-over and two-under” conventional braided weave.

3.7.2.2 Type II. The outer braid of the Type II cord shall be braided over the core (see 3.7.1.2) in a basket weave construction with carriers braiding in pairs.

3.7.2.3 Type III. The outer braid of the Type III cord shall be braided over the core (see 3.7.1.3) so as to give a “two-over and two-under” conventional braided weave.

3.7.3 Length and tips (Type II only). Unless otherwise specified, the lacing cord (Type II) shall be 24 (± 1) inches in length. The ends of the cord shall be tipped with nylon sheets or film,

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or may be tipped or impregnated with cellulose acetate or cellulose acetate butyrate to prevent raveling. The tipped or impregnated end shall be at least 1/2 inch in length.

3.8 Length and put-up. (For Government procurements only.)

3.8.1 Type I and Type III. Unless otherwise specified, the Type I and Type III cord shall be put up on flat head type spools, containing 144 (± 2) yards, with not more than three (3) pieces and no piece measuring less than 10 yards in length. The cord shall be wound on the spools such that each turn and layer is free from entanglement.

3.8.2 Type II. Unless otherwise specified, one gross (144) of Type II lacing cords shall be securely grouped/bundled full length. Cotton tape or twine may be used, providing the tape or twine does not cause damage to, or leave residue or color on, the cords.

3.9 Identification. Unless otherwise specified in the contract or purchase order, each roll or bundle of finished cord shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act. The label shall be such that it remains attached to the spool or bundle until the cord on the spool or in the bundle is fully expended.

3.10 Workmanship. The finished cords shall conform to the quality of product established by this specification. The cords shall be free from all defects (see Table III) which would affect serviceability, appearance and uniformity.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.1.1 Inspection conditions. Unless otherwise excluded, amended, modified or qualified in this specification, or in the contract or purchase order, inspections shall be conducted in accordance with all requirements of the referenced documents and tested in accordance with Tables IV and V.

4.2 First article inspection. When a first article is required (see 3.1 and 6.3), a first article sample shall be submitted to the examinations of 4.4 and the tests of 4.5, as applicable. The presence of excessive defects (see Table III), as defined in the contract (see 6.2), or failure to pass any test shall be cause for rejection of the first article. The sample unit shall be one (1) linear yard for Type I cord and one (1) tipped lacing cord for Type II. The lot size shall be expressed in linear yards for Type I cord; or lacing cords with tips for Type II.

4.2.1 First article sample sizes. Unless otherwise specified in the contract or purchase order, sample sizes for first article inspection shall be in accordance with Table II.

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TABLE II. Sample sizes.

Lot size (see 4.2)	Sample size (sample units)
Up to 1,200	3
1,201 up to and including 3,200	5
3,201 up to and including 10,000	8
10,001 up to and including 35,000	13
35,001 up to and including 150,000	20
150,001 or more	32

4.3 Conformance inspection. Finished cords shall be inspected in accordance with 4.3.1. Except where otherwise indicated (see 6.2), sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 (see 2.3) and with acceptance quality limits (AQL) as specified in the contract and/or order.

4.3.1 End-item inspection. Unless otherwise excluded, amended, modified or qualified in this specification, or in the contract or purchase order, end-item inspection shall include the examinations of paragraphs 4.4 and the tests of 4.5, as applicable. The sample unit shall be one (1) linear yard for Type I cord and one (1) tipped lacing cord for Type II. The lot size shall be expressed in linear yards for Type I cord; or lacing cords with tips for Type II.

4.4 Visual examinations. The cords shall be visually examined for the applicable defects listed in Table IV of MIL-STD-3064. All end-item defects which are clearly noticeable at normal viewing distance (3 feet) and which affect serviceability and appearance of the finished cords shall be scored. All defects found shall be counted, regardless of their proximity to each other. A continuous defect in Type I cords shall be counted as one defect for each linear yard or fraction thereof in which it occurs. In addition to the defects listed in Table IV of MIL-STD-3064, the defects specified in Table III below shall be scored four (4) demerit points per occurrence. Acceptance criteria shall be as specified in the contract or order.

TABLE III. End item visual defects.

Core not completely covered by braiding
Any cut that does not separate the cord into separate pieces
Broken or missing end
Any grease, oil or other substance that transfers to a clean cloth when rubbed
Any spot or stain visible at a normal viewing distance
Uneven braiding, resulting in soft spot
Mottled or uneven color

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4.5 Testing.

4.5.1 Component and material testing. Yarns and elastic strands shall be tested in accordance with Table IV to ensure compliance with the requirements of 3.5 and Table I.

TABLE IV. Component test methods.

Component	Characteristic	Requirement Paragraph	Test Method
Nylon or polyester yarn	Material, identification – Cover, construction Core (singles, Type II) Number of yarns Direction of twist	3.5.1 3.7.2 3.7.1.2 3.7.1.2 3.7.1.2	AATCC-20 1/ Visual Visual Visual ASTM D1423
Elastic strands	Material, identification 1/ Thickness (gauge)	3.5.2 Table I	AATCC-20 1/ 4.6.2

1/ See 6.5.

4.5.2 End-item testing. Finished cords shall be tested for the characteristics specified in Table V. The lot size shall be expressed in linear yards for Type I cord; or lacing cords with tips for Type II. The sample unit for Type I shall be a continuous piece of seven (7) linear yards. The sample unit for Type II shall be ten (10) lacing cords, properly tipped. The sample size shall be in accordance with the following.

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

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TABLE V. End item test methods.

Characteristic	Requirement paragraph	Test method
Shade matching	3.2	Visual
Colorfastness to: Laundering Weathering Crocking Dry/Wet	 3.6.2 3.6.2 3.6.2	 AATCC Test Method 61 Test 1A 1/ 2/ AATCC Test Method 192 1/ 3/ AATCC Test Method 8 4/
End item dimensions Diameter Cord length Type I Type II	 Table I 3.8.1 3.7.3	 4.6.1
Weight per linear yard	Table I	ASTM D3776/D3776M
Total number of elastic strands	Table I	Visual
Number of carriers	Table I	Visual
Ends per carrier	Table I	Visual
Picks per inch	Table I	4.6.3
Elongation	Table I	4.6.4
Weave	3.7.2	Visual

1/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.
2/ Rated using the AATCC Evaluation Procedure 2, Gray Scale for Staining.
3/ Time of exposure shall be 40 AATCC standard fading hours.
4/ Rated using the AATCC Evaluation Procedure 8, 9-Step Chromatic Transference Scale.

4.6 Methods of inspection.

4.6.1 End item dimensional examination. All cords shall be measured in the relaxed state. Any appropriate calibrated measuring device may be used to obtain measurements.

4.6.1.1 Type I cord length. Each spool of Type I cord shall be examined for total length compared to markings. Any spool with a length that is more than one (1) linear yard less than the marked length shall be cause for rejection of the lot. If the total length of the lot is more than two (2) yards less than the total in the contract or purchase order, the lot shall be rejected.

4.6.1.2 Type II lacing cord length. Each bundle of Type II lacing cords shall also be examined for total length compared to the requirements in 3.7.3. Any cord with a length that is more than 1 inch less than the required length shall be cause for rejection of the lot.

4.6.2 Gauge of elastic strands. The gauge of the rubber (elastic strands) shall be determined by counting the actual number of strands, laid side to side, contained in one (1) inch. The gauge is equivalent to the actual number of strands contained in one inch. A calibrated measuring

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device capable of accurately measuring the gauge of the individual rubber strands to an accuracy of 1/10,000 inch may be used.

4.6.3 Picks per inch. Picks per inch shall be counted visually with the cords so positioned that the repeat of the stitch pattern creates a straight line parallel to the axis of the cord.

4.6.4 Elongation. A 14-inch specimen shall be cut from the sample. Two (2) marks shall be made on the cord so that the distance of 10 (±1/32) inches is between the marks. The cord shall be suspended from a clamp in such a manner to allow a 4-pound weight to be hung on the lower end of a Type I cord, and a 2-pound weight to be hung on the lower end of a Type II cord. Gradually lower the weight until the entire load is carried by the cord. After two (2) minutes, take a measurement between the two marks and calculate the increase in length as follows:

$$\frac{B-A}{A} \times 100 = \text{percent elongation}$$

Where
A = initial measurement
B= measurement after 2 minutes

4.7 Visual shade matching. The color and appearance of the finished cord shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, with sources stimulating artificial daylight D75 illuminant with a color temperature of 7500 K (±200), with illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (±200)..

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cords are used in equipment items used by the US armed forces where elasticity is desired. The cords are sometimes considered to be part of the family of items referred to as "shock cord".

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2 and 2.3).
d. When first article sample is required (see 3.1).
e. Number of first article samples (see 4.2).
f. First article and conformance inspection acceptance quality limits (see 4.3, 4.5 and 4.6).
g. Inspection conditions (see 4.1.1).
h. Packaging (see 5.1).

6.3 First article. When a first article is required (see 3.1), it should consist of pre-production samples representative of the quality of production samples. The contracting officer should specify the appropriate type, size and quantity of first article units to be furnished. The contracting officer should include specific instructions in all acquisitions documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Standard sample. For access to standard shade samples and government patterns, address the contracting activity.

6.5 Certificates of compliance. The contracting activity may select to accept a certificate of compliance for stated requirements.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Subject term (key word) listing.

Elongation
Lacing

Custodian:
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA – CT

(Project: 8305-2019-022)

Reviewing activities:
Army - MD

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